

# VoltStream Energy Monitoring Platform

## Complete Knowledge Base Documentation

Version: 3.0.1 | Date: May 2026 | Classification: Customer-Facing

### PAGE 1 — Billing & Subscription

#### 1.0 Subscription Plans & Pricing

VoltStream offers 4 subscription tiers. Annual billing saves exactly 20% versus monthly.

| Plan       | Monthly | Annual  | Devices   | Locations | Storage |
|------------|---------|---------|-----------|-----------|---------|
| Starter    | ₹499    | ₹4,790  | 3         | 1         | 2 GB    |
| Pro        | ₹1,299  | ₹12,470 | 15        | 1         | 20 GB   |
| Business   | ₹4,999  | ₹47,990 | 100       | 5         | 200 GB  |
| Enterprise | Custom  | Custom  | Unlimited | Unlimited | Custom  |

All plans are subject to **18% GST** on the base amount. A Starter Plan subscriber in Hyderabad pays ₹499 + ₹89.82 GST = **₹588.82/month total**. A Pro Plan subscriber pays ₹1,299 + ₹233.82 = **₹1,532.82/month total**.

#### 1.1 Proration Formula

When a user upgrades mid-cycle:

**Prorated Charge = (New Daily Rate – Old Daily Rate) × Remaining Days**

**Example:** User upgrades from Starter (₹499) to Pro (₹1,299) on Day 20 of a 30-day cycle.

- Starter daily rate = ₹499 ÷ 30 = **₹16.63/day**
- Pro daily rate = ₹1,299 ÷ 30 = **₹43.30/day**
- Remaining days = 10
- Prorated charge = (₹43.30 – ₹16.63) × 10 = **₹266.70**

From the next cycle onward, the full ₹1,299/month applies.

#### 1.2 Annual Refund Formula

**Refund = (Annual Fee ÷ 12) × Remaining Full Months**

**Example:** User paid ₹12,470 for annual Pro Plan, cancels after 4 months.

- Monthly equivalent = ₹12,470 ÷ 12 = **₹1,039.17**
- Remaining months = 8

- Refund = ₹1,039.17 × 8 = **₹8,313.36**
- Processed within **7–14 business days**

---

### 1.3 Device Overage Billing

Business Plan allows 100 devices. Beyond that, **₹49/device/month** is charged.

**Example:** Business subscriber connects 118 devices in June.

- Overage devices = 118 – 100 = **18 devices**
- Overage charge = 18 × ₹49 = **₹882/month**
- Total bill = ₹4,999 + ₹882 = **₹5,881 + GST = ₹6,939.58**

---

### 1.4 Failed Payment Retry Schedule

| Retry Attempt       | Timing After Failure | Account Status |
|---------------------|----------------------|----------------|
| Attempt 1           | 24 hours             | Fully Active   |
| Attempt 2           | 48 hours (Day 3)     | Fully Active   |
| Attempt 3           | 6 days (Day 7)       | Fully Active   |
| Suspension          | Day 8                | Restricted     |
| Termination Warning | Day 30               | Final Notice   |
| Data Deletion       | Day 37               | Permanent      |

Data continues to be collected passively for **14 days post-suspension**. After termination at Day 37, all account data is permanently deleted.

---

### 1.5 Billing FAQ

**Q: A Business Plan user had 125 devices active for 30 days. What is the total monthly invoice including GST?** Base = ₹4,999. Overage = 25 × ₹49 = ₹1,225. Subtotal = ₹6,224. GST 18% = ₹1,120.32. **Total = ₹7,344.32**

**Q: How many days of energy data does a Pro Plan subscriber retain?** Pro Plan retains **24 months = 730 days** of usage history.

**Q: What is the annual saving if a Business Plan user switches from monthly to annual billing?** Monthly total = ₹4,999 × 12 = ₹59,988. Annual price = ₹47,990. **Saving = ₹11,998/year**

## 2.0 Supported Device Categories

VoltStream supports the following IoT and smart energy device types:

| Device Type        | Max per Starter | Max per Pro | Max per Business |
|--------------------|-----------------|-------------|------------------|
| Smart Meters       | 1               | 5           | 30               |
| IoT Energy Sensors | 1               | 6           | 40               |
| Smart Plugs        | 1               | 4           | 20               |
| EV Chargers        | 0               | 2           | 10               |
| Solar Inverters    | 0               | 2           | 10               |
| HVAC Monitors      | 0               | 1           | 10               |

### 2.1 Smart Meter Setup Process

Setting up a smart meter on VoltStream takes an average of **12 minutes** for a first-time user. The process follows 6 steps:

**Step 1 — Hardware Registration (2 min):** Locate the 16-digit Device ID printed on the back of the meter (format: VS-XXXX-XXXX-XXXX). Enter this in the VoltStream app under **Devices → Add New → Smart Meter**.

**Step 2 — Wi-Fi Pairing (3 min):** The meter supports 2.4 GHz Wi-Fi only (not 5 GHz). Ensure the router is within **15 meters** of the meter during setup. The meter uses **WPA2/WPA3** security protocols.

**Step 3 — Utility Integration (2 min):** Select your electricity provider from the list of **47 supported DISCOMs** in India. Enter your Consumer Number to enable automatic tariff mapping.

**Step 4 — Tariff Configuration (2 min):** Choose from Flat Rate, Time-of-Use (ToU), or Slab tariff structures. Enter your per-unit rate. Example: TSSPDCL residential users in Hyderabad pay ₹2.35/unit for 0–100 kWh, ₹4.80/unit for 101–300 kWh, and ₹7.50/unit above 300 kWh.

**Step 5 — Calibration (2 min):** The meter performs an automatic baseline calibration reading over **60 seconds**. It measures ambient current draw with a precision of **±0.5%**.

**Step 6 — First Reading (1 min):** The first energy data point appears in the dashboard within **90 seconds** of successful calibration.

---

### 2.2 Firmware Updates

VoltStream pushes firmware updates Over-the-Air (OTA) to all connected devices. Update frequency averages **once every 6 weeks**. Each firmware package is typically **2.4 MB to 6.8 MB** in size. Updates are pushed between **2:00 AM and 4:00 AM local time** to minimize disruption.

During a firmware update, the device goes offline for approximately **3–7 minutes**. No energy data is lost during this window — the device buffers up to **72 hours** of readings locally and syncs them once it reconnects.

If a firmware update fails, the device automatically rolls back to the last stable version. VoltStream retains the previous **3 firmware versions** for rollback compatibility. Users can check device firmware version under **Devices → [Device Name] → Info → Firmware Version**.

---

## 2.3 Troubleshooting Offline Devices

A device is marked **Offline** if it has not sent a data packet in more than **5 minutes** (Pro/Business) or **10 minutes** (Starter). The dashboard displays a red indicator and triggers an alert if the device has been offline for more than **15 consecutive minutes**.

| Issue                  | Likely Cause                | Resolution Time           |
|------------------------|-----------------------------|---------------------------|
| No Wi-Fi signal        | Router out of range (>15m)  | 2 minutes                 |
| IP conflict            | Duplicate device on network | 5 minutes                 |
| Failed firmware update | Corrupt OTA package         | Auto-resolved in 10 min   |
| Power outage           | Mains disconnected          | Resolves on power restore |
| Server sync failure    | Backend queue delay         | Auto-resolved in <30 min  |

**Energy Sensor Calibration:** Sensors should be recalibrated every **180 days**. Calibration drift beyond **±2%** triggers an automatic alert. Manual recalibration takes **5 minutes** and requires the device to be idle (0 load) for the first **30 seconds** of the process.

---

## 2.4 Smart Device FAQ

**Q: A Business Plan user has 3 smart meters, 12 IoT sensors, 8 smart plugs, 4 EV chargers, and 5 solar inverters. How many total devices is that, and are they within plan limits?** Total = 3 + 12 + 8 + 4 + 5 = **32 devices**. Business Plan limit is 100. **Within limit — no overage.**

**Q: How long does a smart meter buffer data during a firmware update?** Up to **72 hours** of readings are stored locally and synced after reconnection.

**Q: What is the calibration accuracy tolerance for energy sensors?** Sensors operate with **±0.5% precision** at setup. Recalibration is triggered if drift exceeds **±2%**.

---

## PAGE 3 — Analytics Dashboard

### 3.0 Dashboard Overview

The VoltStream Analytics Dashboard is the central interface for monitoring, analyzing, and acting on energy data. It aggregates readings from all connected devices and presents them across 5 primary view modules: **Real-Time Monitor**, **Consumption Analytics**, **Billing Estimate**, **Peak Hour Analysis**, and **Usage Predictions**.

---

### 3.1 Real-Time Monitoring Refresh Rates

Data refresh rates differ by plan:

| Plan       | Refresh Interval | Daily Data Points per Device |
|------------|------------------|------------------------------|
| Starter    | 5 minutes        | 288                          |
| Pro        | 1 minute         | 1,440                        |
| Business   | 30 seconds       | 2,880                        |
| Enterprise | 10 seconds       | 8,640                        |

A Pro Plan user with 10 devices generates  $1,440 \times 10 = 14,400$  data points per day. Over 30 days that is **432,000 data points/month** stored per account.

---

### 3.2 Energy Consumption Analytics

The consumption analytics module calculates:

**Daily Average Consumption (kWh)** = Total Monthly kWh  $\div$  Days in Month

**Example:** A household consumes 420 kWh in the month of June (30 days).

- Daily average =  $420 \div 30 = 14$  kWh/day
- Hourly average =  $14 \div 24 = 0.583$  kWh/hour
- Monthly cost at flat ₹7.50/unit =  $420 \times ₹7.50 = ₹3,150$

The dashboard also computes a **Month-over-Month (MoM) change**:

**MoM Change %** =  $((\text{Current Month kWh} - \text{Previous Month kWh}) \div \text{Previous Month kWh}) \times 100$

**Example:** May consumption = 380 kWh, June consumption = 420 kWh.

- MoM Change =  $((420 - 380) \div 380) \times 100 = +10.53\%$  increase
- 

### 3.3 Peak Hour Monitoring

VoltStream defines **peak hours** as the period between **6:00 PM and 10:00 PM** local time, when grid demand and ToU tariff rates are at their highest. The dashboard tracks:

- Peak hour consumption as a % of total daily consumption
- Cost incurred during peak hours vs off-peak hours
- Top 3 energy-consuming devices during peak windows

**Example:** A household uses 14 kWh/day. 5.6 kWh (40%) is consumed during peak hours (6–10 PM). Using a ToU tariff of ₹9/unit (peak) vs ₹6/unit (off-peak):

- Peak cost =  $5.6 \times ₹9 = ₹50.40$
- Off-peak cost =  $8.4 \times ₹6 = ₹50.40$

- Total daily cost = ₹100.80
- If all 14 kWh were shifted to off-peak:  $14 \times ₹6 = ₹84.00$  — saving ₹16.80/day = ₹504/month

---

### 3.4 Usage Predictions

The Pro Plan and above include a **7-day energy forecast** powered by VoltStream's ML prediction engine. The model is trained on:

- **90 days** of historical consumption data
- Weather API data (temperature, humidity)
- Day-of-week usage patterns
- Historical appliance runtime logs

Prediction accuracy averages **±4.2% error margin** across residential users. For commercial users with more consistent usage patterns, accuracy improves to **±2.1%**.

#### Example Forecast Output:

| Day       | Predicted kWh | Confidence |
|-----------|---------------|------------|
| Monday    | 13.8          | 96%        |
| Tuesday   | 14.1          | 95%        |
| Wednesday | 15.6          | 91%        |
| Thursday  | 16.2          | 89%        |
| Friday    | 17.4          | 87%        |
| Saturday  | 19.1          | 84%        |
| Sunday    | 18.3          | 85%        |

### 3.5 Dashboard FAQ

**Q: A Pro user has 8 devices. How many total data points are stored in 30 days?** 1,440 points/device/day  $\times$  8 devices  $\times$  30 days = **345,600 data points**

**Q: A household consumed 510 kWh in July and 460 kWh in June. What is the MoM change?**  $((510 - 460) \div 460) \times 100 =$  **+10.87% increase**

**Q: If 45% of daily consumption (18 kWh total) falls in peak hours at ₹9/unit, and the rest is at ₹6/unit, what is the daily cost?** Peak =  $18 \times 0.45 \times ₹9 = ₹72.90$ . Off-peak =  $18 \times 0.55 \times ₹6 = ₹59.40$ . Total = **₹132.30/day**

## 4.0 How VoltStream Generates Recommendations

VoltStream's recommendation engine analyzes **90 days of consumption data**, appliance runtime logs, and local tariff schedules to generate personalized energy-saving actions. Recommendations are ranked by **estimated monthly savings in ₹**, highest first. Each recommendation includes a confidence score based on data quality and model certainty.

Recommendations are refreshed every **24 hours** and are accessible under **Dashboard → Recommendations**. Pro and Business users receive up to **15 active recommendations** at any time, while Starter users receive up to **5**.

---

### 4.1 High-Impact Appliance Optimization

**Air Conditioners** are the largest energy consumers in most Indian households, accounting for **40–55% of total monthly electricity usage** during summer months. VoltStream tracks AC runtime, setpoint temperature, and ambient temperature to compute savings potential.

**Key insight:** Increasing AC setpoint from 18°C to 24°C reduces energy consumption by approximately **6% per degree**, per BEE (Bureau of Energy Efficiency) standards.

**Example:** An AC rated at **1.5 kW** runs for **8 hours/day** at 18°C.

- Daily consumption =  $1.5 \times 8 = \mathbf{12\ kWh}$
- Monthly =  $12 \times 30 = \mathbf{360\ kWh}$
- Cost at ₹7.50/unit = **₹2,700/month**

If setpoint is changed to 24°C (6°C difference):

- Savings =  $6\% \times 6 = \mathbf{36\%\ reduction}$
  - New consumption =  $360 \times (1 - 0.36) = \mathbf{230.4\ kWh}$
  - New cost =  $230.4 \times ₹7.50 = \mathbf{₹1,728/month}$
  - **Monthly saving = ₹972**
- 

### 4.2 Smart Scheduling

VoltStream's smart scheduling feature allows users to automate high-wattage appliances to run during **off-peak hours (10 PM–6 AM)** where ToU tariffs are lower.

**Recommended devices for off-peak scheduling:**

- Washing machines (average draw: **500W–2,000W**)
- Dishwashers (average draw: **1,200W–1,800W**)
- Water heaters/geysers (average draw: **2,000W–3,000W**)
- EV chargers (average draw: **3,300W–7,200W**)

**Example — EV Charging Savings:** An EV charger draws **7.2 kW** and charges for **2 hours/night**.

- Daily consumption =  $7.2 \times 2 = \mathbf{14.4\text{ kWh}}$
- At peak tariff (₹9/unit) =  $14.4 \times ₹9 = \mathbf{₹129.60/\text{day}}$
- At off-peak tariff (₹6/unit) =  $14.4 \times ₹6 = \mathbf{₹86.40/\text{day}}$
- **Daily saving = ₹43.20 | Monthly saving = ₹1,296**

---

### 4.3 Standby Power Waste Detection

VoltStream monitors devices for **standby power draw** — energy consumed while appliances are plugged in but not actively in use. The average Indian household wastes **65–80 kWh/month** on standby power, costing approximately **₹487–₹600/month** at ₹7.50/unit.

Common standby culprits detected by VoltStream:

| Appliance       | Standby Wattage | Monthly Standby kWh | Monthly Cost |
|-----------------|-----------------|---------------------|--------------|
| Television      | 5W              | 3.6 kWh             | ₹27          |
| Microwave       | 3W              | 2.16 kWh            | ₹16.20       |
| Set-top Box     | 15W             | 10.8 kWh            | ₹81          |
| Desktop PC      | 6W              | 4.32 kWh            | ₹32.40       |
| Washing Machine | 2W              | 1.44 kWh            | ₹10.80       |

Eliminating standby waste for all 5 devices above saves **22.32 kWh/month = ₹167.40/month** at ₹7.50/unit.

---

### 4.4 Sustainability Metrics

VoltStream calculates your **carbon footprint** based on energy consumption using India's national grid emission factor of **0.82 kg CO<sub>2</sub> per kWh** (Central Electricity Authority, 2025).

**Example:** A household consumes 420 kWh/month.

- CO<sub>2</sub> emissions =  $420 \times 0.82 = \mathbf{344.4\text{ kg CO}_2/\text{month}}$
- Annual emissions =  $344.4 \times 12 = \mathbf{4,132.8\text{ kg CO}_2/\text{year} = 4.13\text{ tonnes CO}_2}$

If recommendations reduce consumption by 20%:

- New monthly consumption =  $420 \times 0.80 = \mathbf{336\text{ kWh}}$
- CO<sub>2</sub> saved =  $(420 - 336) \times 0.82 = \mathbf{68.88\text{ kg CO}_2/\text{month}}$
- Annual CO<sub>2</sub> saving =  $\mathbf{826.56\text{ kg} = 0.83\text{ tonnes/year}}$

---

### 4.5 Energy Saving FAQ



**Q: An AC (2 kW) runs 10 hours/day at 16°C. What is monthly cost at ₹8/unit and how much is saved by raising to 24°C?** Consumption =  $2 \times 10 \times 30 = 600$  kWh. Cost =  $600 \times ₹8 = ₹4,800$ . Savings at 8°C increase =  $8 \times 6\% = 48\%$  reduction. New cost =  $₹4,800 \times 0.52 = ₹2,496$ . **Saving = ₹2,304/month**

**Q: What is the annual CO<sub>2</sub> footprint for a household consuming 550 kWh/month?**  $550 \times 0.82 \times 12 = 5,412$  kg = **5.41 tonnes CO<sub>2</sub>/year**

---

## PAGE 5 — User Account & Security

### 5.0 Account Structure

Every VoltStream account has exactly **1 Account Owner** and supports up to:

- **5 Billing Administrators**
- **50 Standard Users** (Business Plan)
- **10 Standard Users** (Pro Plan)
- **2 Standard Users** (Starter Plan)

Enterprise accounts support **unlimited users** with SCIM-based provisioning and SSO (Single Sign-On) via SAML 2.0.

---

### 5.1 Password & Authentication Policy

VoltStream enforces the following password policy:

| Requirement                | Value                                      |
|----------------------------|--------------------------------------------|
| Minimum length             | 10 characters                              |
| Must include               | 1 uppercase, 1 number, 1 special character |
| Maximum age                | 90 days (Business/Enterprise)              |
| History restriction        | Last 5 passwords cannot be reused          |
| Failed login lockout       | After 5 attempts — 15-minute lockout       |
| Session timeout (inactive) | 30 minutes                                 |
| Session timeout (absolute) | 12 hours                                   |

**Multi-Factor Authentication (MFA)** is optional for Starter/Pro users and **mandatory** for Business/Enterprise users. Supported MFA methods: TOTP apps (Google Authenticator, Authy), SMS OTP, and hardware keys (FIDO2/WebAuthn).

---

### 5.2 Multi-Device Login

VoltStream allows concurrent sessions across multiple devices:

| Plan       | Max Concurrent Sessions |
|------------|-------------------------|
| Starter    | 2                       |
| Pro        | 5                       |
| Business   | 15                      |
| Enterprise | Unlimited               |

Each session is tracked with a unique session token with a **128-bit entropy**. When a user exceeds their session limit, the oldest session is automatically invalidated. Users can view and manually revoke active sessions under **Settings → Security → Active Sessions**.

---

### 5.3 Role-Based Access Control (RBAC)

VoltStream uses a 4-tier RBAC model:

| Role          | Billing Access | Device Management     | Analytics | User Management |
|---------------|----------------|-----------------------|-----------|-----------------|
| Account Owner | Full           | Full                  | Full      | Full            |
| Billing Admin | Full           | Read-only             | Full      | None            |
| Site Manager  | None           | Full (assigned sites) | Full      | Limited         |
| Standard User | None           | Read-only             | View only | None            |

Business Plan users can create **custom roles** with granular permission sets. Enterprise clients can define up to **20 custom roles** per account.

---

### 5.4 Data Privacy & Retention

VoltStream complies with **India's Digital Personal Data Protection Act (DPDPA) 2023** and GDPR for EU users. Key data handling metrics:

| Data Type          | Retention Period                | Encryption        |
|--------------------|---------------------------------|-------------------|
| Energy usage data  | Per plan (6 months–5 years)     | AES-256 at rest   |
| Login/session logs | 12 months                       | AES-256 at rest   |
| Payment records    | 7 years (GST compliance)        | AES-256 + TLS 1.3 |
| Device telemetry   | 90 days raw, aggregated forever | AES-256 at rest   |
| Support tickets    | 3 years                         | AES-256 at rest   |

All data is stored in **AWS Mumbai (ap-south-1)** region for Indian accounts. International accounts use **AWS Frankfurt (eu-central-1)** or **AWS Oregon (us-west-2)** based on user geography.

Data deletion requests under DPDPA are processed within **30 days** of request submission. Users can submit a deletion request at **Settings → Privacy → Request Data Deletion**.

---

## 5.5 Notification Channels & Alert Thresholds

VoltStream supports 3 notification channels: **Email, SMS, and In-App Push**. Users can configure the following alert types:

| Alert Type                 | Default Threshold         | Customizable?  |
|----------------------------|---------------------------|----------------|
| Daily consumption exceeded | 20 kWh                    | Yes            |
| Monthly budget exceeded    | ₹3,000                    | Yes            |
| Device offline             | 15 minutes                | Yes (5–60 min) |
| Failed payment             | Immediate                 | No             |
| Unusual spike detected     | +50% above 7-day avg      | Yes            |
| Firmware update available  | Immediate                 | No             |
| Peak hour warning          | 30 min before peak window | Yes            |

SMS alerts are charged at **₹0.15/SMS** beyond the monthly free quota. Free SMS quota per plan: Starter = **50 SMS/month**, Pro = **200 SMS/month**, Business = **1,000 SMS/month**.

**Example:** A Pro user sends 245 SMS alerts in a month.

- Free quota = 200 SMS
- Overage = 45 SMS × ₹0.15 = **₹6.75** added to next invoice

---

## 5.6 Account & Security FAQ

**Q: A Business Plan account has 14 active sessions. A 15th login occurs. What happens?** The 15th login is permitted (Business limit = 15). The **oldest session is invalidated** automatically if a 16th login is attempted.

**Q: A Pro user sends 310 SMS notifications in a month. What is the overage charge?** Free quota = 200. Overage = 110 × ₹0.15 = **₹16.50**

**Q: How long does VoltStream retain payment records and under which compliance regulation?** **7 years** under **GST compliance** requirements. Encrypted with AES-256 + TLS 1.3.

---

VoltBot is VoltStream's built-in AI assistant, powered by a **Retrieval-Augmented Generation (RAG)** architecture. It combines a large language model with a real-time knowledge retrieval system that searches across VoltStream's documentation, the user's own energy data, and uploaded custom documents to provide contextually accurate answers.

VoltBot is accessible from the bottom-right chat icon on the dashboard, available **24/7**, and supports queries in **English, Hindi, and Telugu** as of version 3.0.

---

### 6.1 VoltBot Usage Quotas

| Plan       | Monthly Queries | Response Context Window | Custom Doc Upload    |
|------------|-----------------|-------------------------|----------------------|
| Starter    | 30              | 4,000 tokens            | No                   |
| Pro        | 300             | 8,000 tokens            | No                   |
| Business   | Unlimited       | 16,000 tokens           | Yes (up to 20 docs)  |
| Enterprise | Unlimited       | 32,000 tokens           | Yes (unlimited docs) |

Each query consumes **1 credit** regardless of response length. Queries that trigger document retrieval (RAG lookups) consume **2 credits** on Starter and Pro plans. If a Pro user exhausts their 300 monthly credits, additional packs of **100 credits cost ₹199/pack**.

---

### 6.2 How RAG Works in VoltBot

VoltBot's RAG pipeline operates in **4 stages** with the following average processing times:

| Stage        | Process                                         | Avg. Time       |
|--------------|-------------------------------------------------|-----------------|
| 1            | Query embedding generation                      | 120ms           |
| 2            | Vector similarity search across knowledge base  | 85ms            |
| 3            | Top-K document chunk retrieval (K=5 by default) | 40ms            |
| 4            | LLM response generation with retrieved context  | 1,800ms         |
| <b>Total</b> | End-to-end response                             | <b>~2,045ms</b> |

The knowledge base is indexed using **1,536-dimensional vector embeddings**. Document chunks are split at **512 tokens** with a **64-token overlap** to preserve contextual continuity across chunk boundaries. The similarity search uses **cosine similarity** with a minimum relevance threshold of **0.72** — chunks scoring below this are excluded from the context even if they rank in the top-K results.

---

### 6.3 Supported Question Types

VoltBot can answer questions across 6 domains:

- 1. Billing & Invoice Queries:** "What is my current outstanding invoice?", "Why was I charged ₹882 extra this month?", "When is my next renewal date?"
- 2. Device & Sensor Status:** "Is my smart meter offline?", "Which device has the highest consumption today?", "When was the last firmware update applied to Sensor-04?"
- 3. Consumption Analytics:** "What was my average daily usage last month?", "Compare my June and July consumption.", "Which hour of the day has my peak usage?"
- 4. Energy Saving Advice:** "How can I reduce my electricity bill this summer?", "What appliances are running during peak hours?", "How much would I save by shifting my washing machine to night hours?"
- 5. Plan & Subscription Guidance:** "What plan should I upgrade to if I want to add 8 more devices?", "How much will I be charged if I upgrade today?", "What is the refund for my annual plan if I cancel now?"
- 6. Document-Grounded Q&A (Business/Enterprise):** VoltBot can answer questions from uploaded documents such as equipment manuals, utility tariff sheets, or internal energy policies. Example: "According to the TSSPDCL tariff sheet I uploaded, what is the per-unit rate for consumption above 500 units?"

---

### 6.4 VoltBot Limitations

VoltBot has defined operational boundaries users should be aware of:

- VoltBot **cannot execute actions** on behalf of users — it can advise on how to change a setting but cannot apply the change itself (as of v3.0; action mode is on the roadmap for v3.5, targeted for Q3 2026).
- VoltBot **cannot access data older than the plan's retention limit**. A Starter user asking about data from 9 months ago will receive a partial or null response.
- VoltBot **does not provide legal or regulatory advice** about electricity regulations, tariff disputes with utilities, or contractual matters.
- VoltBot responses carry a **disclaimer** when answering billing estimate questions, noting that figures are approximate and may differ from official utility invoices.
- VoltBot's knowledge base is updated every **7 days**. Information about very recent platform changes (within 7 days of a release) may not yet be reflected.

---

### 6.5 VoltBot Accuracy & Performance Benchmarks

VoltStream publishes the following internal benchmarks for VoltBot (based on testing across 10,000 user queries, Q1 2026):

| Query Category  | Accuracy Rate | Avg. Response Time |
|-----------------|---------------|--------------------|
| Billing queries | 97.3%         | 1.9 seconds        |

| Query Category        | Accuracy Rate | Avg. Response Time |
|-----------------------|---------------|--------------------|
| Device status         | 99.1%         | 1.4 seconds        |
| Consumption analytics | 95.8%         | 2.1 seconds        |
| Energy saving advice  | 91.4%         | 2.4 seconds        |
| Document-grounded Q&A | 88.6%         | 3.2 seconds        |
| Plan recommendations  | 94.2%         | 1.8 seconds        |

Overall average accuracy across all categories: **94.4%**. Overall average response time: **2.1 seconds**.

---

## 6.6 VoltBot FAQ

**Q: A Pro user has used 285 queries so far this month. They need 40 more. What will they be charged?** Remaining free =  $300 - 285 = 15$ . Overage =  $40 - 15 = 25$  queries. Minimum pack = 100 queries for ₹199. **Charge = ₹199** (only 1 pack needed, 75 credits left over for the month).

**Q: What is the minimum cosine similarity score for a document chunk to be included in a VoltBot RAG response?** **0.72**. Chunks scoring below this threshold are excluded regardless of ranking.

**Q: How often is VoltBot's knowledge base updated?** Every **7 days**.

---

## PAGE 7 — Tariff Structures & Energy Cost Calculations

### 7.0 Overview of Tariff Types

VoltStream supports 4 tariff structure types that users can configure under **Settings → Energy Settings → Tariff Configuration**. The correct tariff setup is critical for accurate billing estimates and savings recommendations.

---

#### 7.1 Flat Rate Tariff

A single per-unit rate applies at all times, regardless of time of day or consumption volume.

**Example — BESCO (Bangalore) Residential Flat Rate:**

- Rate: ₹5.70/kWh for all consumption above 200 units

**Calculation:** Household uses 350 kWh in August.

- First 200 kWh at lower slab (₹3.15):  $200 \times ₹3.15 = ₹630$
- Next 150 kWh at ₹5.70:  $150 \times ₹5.70 = ₹855$
- Fixed charge: ₹75
- **Total estimated bill = ₹1,560**

---

## 7.2 Time-of-Use (ToU) Tariff

Different rates apply during defined time windows. VoltStream maps every consumption reading to the correct rate band using the device timestamp.

### Example — Commercial ToU Tariff (Hyderabad):

| Time Band | Hours                | Rate       |
|-----------|----------------------|------------|
| Off-Peak  | 10 PM – 6 AM (8 hrs) | ₹6.00/kWh  |
| Standard  | 6 AM – 6 PM (12 hrs) | ₹8.50/kWh  |
| Peak      | 6 PM – 10 PM (4 hrs) | ₹11.00/kWh |

**Monthly bill calculation for a small office using 800 kWh/month:** Assume consumption split: 20% off-peak (160 kWh), 55% standard (440 kWh), 25% peak (200 kWh).

- Off-peak:  $160 \times ₹6.00 = ₹960$
- Standard:  $440 \times ₹8.50 = ₹3,740$
- Peak:  $200 \times ₹11.00 = ₹2,200$
- **Total energy cost = ₹6,900/month**

If the office shifts 100 kWh from peak to off-peak through smart scheduling:

- New peak:  $100 \times ₹11.00 = ₹1,100$
- New off-peak:  $260 \times ₹6.00 = ₹1,560$
- **New total = ₹6,400 — saving ₹500/month**

---

## 7.3 Slab/Block Tariff

The per-unit rate increases as consumption crosses defined thresholds within the billing month.

### Example — TSSPDCL (Telangana) Residential Slab Tariff:

| Slab   | Units         | Rate      |
|--------|---------------|-----------|
| Slab 1 | 0–100 kWh     | ₹2.35/kWh |
| Slab 2 | 101–200 kWh   | ₹4.80/kWh |
| Slab 3 | 201–300 kWh   | ₹6.90/kWh |
| Slab 4 | Above 300 kWh | ₹8.50/kWh |

**Example — Household consuming 380 kWh in July:**

- Slab 1:  $100 \times ₹2.35 = ₹235$
- Slab 2:  $100 \times ₹4.80 = ₹480$

- Slab 3:  $100 \times ₹6.90 = ₹690$
  - Slab 4:  $80 \times ₹8.50 = ₹680$
  - Fixed charges: ₹120
  - **Total = ₹2,205**
- 

#### 7.4 Demand Charge (Commercial)

Commercial tariffs often include a **demand charge** based on peak kW demand recorded during 15-minute intervals.

**Formula:** Demand Charge = Peak Demand (kW) × Demand Rate (₹/kW)

**Example:** A factory records a peak demand of **85 kW** in October. Demand rate = ₹250/kW/month.

- Demand charge =  $85 \times ₹250 = ₹21,250$
- Energy charge ( $60,000 \text{ kWh} \times ₹6.50$ ) = ₹3,90,000
- Fixed charge = ₹5,000
- **Total bill = ₹4,16,250 + GST**

VoltStream tracks demand in **15-minute intervals** and displays the rolling peak demand in real time, alerting users when they approach a demand threshold that would trigger a higher demand charge bracket.

---

#### 7.5 Tariff FAQ

**Q: A household in Telangana consumed 265 kWh in May. Using TSSPDCL slab rates above, what is the energy charge (excluding fixed charges)?** Slab 1:  $100 \times ₹2.35 = ₹235$ . Slab 2:  $100 \times ₹4.80 = ₹480$ . Slab 3:  $65 \times ₹6.90 = ₹448.50$ . **Total energy charge = ₹1,163.50**

**Q: Using the ToU tariff above, if an office shifts its entire 200 kWh peak consumption to off-peak hours, how much does it save monthly?** Original peak cost =  $200 \times ₹11 = ₹2,200$ . Shifted cost =  $200 \times ₹6 = ₹1,200$ . **Monthly saving = ₹1,000**

---

### PAGE 8 — Solar & Renewable Energy Monitoring

#### 8.0 Solar Integration Overview

VoltStream supports monitoring of **rooftop solar PV systems** alongside grid consumption. Users with solar inverters connected to VoltStream get a unified view of energy generation, consumption, grid import, and grid export on a single dashboard. Solar monitoring is available on **Pro Plan and above**.

---

#### 8.1 Key Solar Metrics Tracked



VoltStream tracks the following solar parameters at **1-minute intervals** (Pro Plan):

| Metric                 | Unit | Description                                       |
|------------------------|------|---------------------------------------------------|
| Solar Generation       | kWh  | Total energy produced by panels                   |
| Self-Consumption       | kWh  | Solar energy used directly at the property        |
| Grid Export            | kWh  | Excess solar energy fed back to grid              |
| Grid Import            | kWh  | Energy drawn from grid when solar is insufficient |
| Self-Sufficiency Ratio | %    | Self-consumption ÷ Total consumption              |
| Solar Fraction         | %    | Solar generation ÷ Total consumption              |

## 8.2 Net Metering Calculation

Under net metering, units exported to the grid are credited against grid imports on the monthly bill.

**Net Units Billed = Grid Import (kWh) – Grid Export (kWh)**

**Example:** A household with a **3 kW rooftop solar system** in June:

- Total consumption: 310 kWh
- Solar generation: 270 kWh
- Self-consumption: 200 kWh
- Grid export: 70 kWh
- Grid import: 110 kWh

Net units billed = 110 – 70 = **40 kWh**

At ₹6.90/kWh (Slab 3): 40 × ₹6.90 = **₹276 energy charge** (vs ₹2,139 without solar)

**Monthly saving from solar = ₹2,139 – ₹276 = ₹1,863 Annual saving = ₹1,863 × 12 = ₹22,356**

---

## 8.3 Solar ROI Calculator

VoltStream includes a built-in Solar ROI calculator under **Analytics → Solar → ROI Analysis**.

**Inputs:** System cost, capacity (kW), average daily generation (kWh), applicable tariff, annual degradation rate.

**Standard solar panel degradation rate used by VoltStream: 0.5% per year**

**Example — 3 kW System:**

- System cost: ₹1,80,000
- Average daily generation: 12 kWh
- Annual generation: 12 × 365 = **4,380 kWh**
- Annual saving at ₹7.50/unit: 4,380 × ₹7.50 = **₹32,850/year**

- Simple payback period: ₹1,80,000 ÷ ₹32,850 = **5.48 years**
- 25-year lifetime saving: ₹32,850 × 25 = **₹8,21,250** (before degradation)

After 0.5%/year degradation, Year 10 generation =  $4,380 \times (1 - 0.005)^{10} = 4,380 \times 0.951 = \mathbf{4,165 \text{ kWh}}$  (saving ₹31,238 in Year 10).

## 8.4 Solar FAQ

**Q: A household generates 320 kWh solar, self-consumes 240 kWh, exports 80 kWh, and imports 140 kWh. What are net units billed?** Net units = 140 – 80 = **60 kWh**

**Q: What is the self-sufficiency ratio if total consumption is 350 kWh and self-consumption is 240 kWh?** Self-sufficiency =  $240 \div 350 \times 100 = \mathbf{68.57\%}$

**Q: A 5 kW solar system costs ₹3,00,000 and saves ₹54,000/year. What is the payback period?** ₹3,00,000 ÷ ₹54,000 = **5.56 years**

## PAGE 9 — Platform Performance, SLAs & Integrations

### 9.0 Platform Uptime & SLA

VoltStream publishes monthly uptime reports at [status.voltstream.io](https://status.voltstream.io). SLA commitments by plan:

| Plan       | Uptime SLA | Max Downtime/Month | Max Downtime/Year |
|------------|------------|--------------------|-------------------|
| Starter    | 99.5%      | 3.65 hours         | 43.8 hours        |
| Pro        | 99.7%      | 2.19 hours         | 26.28 hours       |
| Business   | 99.9%      | 43.8 minutes       | 8.76 hours        |
| Enterprise | 99.95%     | 21.9 minutes       | 4.38 hours        |

**Service credits** are issued for downtime exceeding SLA commitments:

| Excess Downtime     | Credit             |
|---------------------|--------------------|
| SLA breach < 1 hour | 5% of monthly fee  |
| 1–4 hours breach    | 15% of monthly fee |
| 4–8 hours breach    | 25% of monthly fee |
| > 8 hours breach    | 50% of monthly fee |

**Example:** A Pro user (₹1,299/month) experiences 5 hours of downtime beyond the SLA limit.

- Credit tier = 25%
- Credit amount = ₹1,299 × 0.25 = **₹324.75** applied to next invoice

## 9.1 API Rate Limits

VoltStream provides a REST API for third-party integrations. Rate limits by plan:

| Plan       | API Calls/min | API Calls/day | Webhook Events/day |
|------------|---------------|---------------|--------------------|
| Starter    | 10            | 1,000         | Not available      |
| Pro        | 60            | 10,000        | 500                |
| Business   | 300           | 1,00,000      | 10,000             |
| Enterprise | Custom        | Unlimited     | Unlimited          |

Exceeding the per-minute rate limit returns HTTP **429 Too Many Requests** with a Retry-After header indicating wait time in seconds. Day-level limits reset at **00:00 UTC** daily.

---

## 9.2 Supported Integrations

VoltStream natively integrates with the following platforms as of v3.0:

| Category           | Integrations                             |
|--------------------|------------------------------------------|
| Smart Home         | Amazon Alexa, Google Home, Apple HomeKit |
| Solar              | SolarEdge, Enphase, Fronius, ABB         |
| EV Charging        | Tata Power EV, Ather Grid, EESL          |
| Utility APIs       | 47 DISCOMs across India                  |
| IFTTT / Automation | IFTTT, Zapier, n8n                       |
| Cloud Storage      | AWS S3, Google Drive, Azure Blob         |
| BI Tools           | Power BI, Tableau, Google Looker         |
| Communication      | Slack, Microsoft Teams, PagerDuty        |

Data exports via API are available in **JSON, CSV, and Parquet** formats. Parquet export is available on Business and Enterprise plans only.

---

## 9.3 Data Ingestion Pipeline

VoltStream's data ingestion pipeline processes readings from smart meters and IoT devices through the following stages:

1. **Device → MQTT Broker** (latency: < 50ms)
2. **MQTT → Stream Processor (Apache Kafka)** (latency: < 100ms)
3. **Kafka → Time-Series Database (InfluxDB)** (latency: < 200ms)
4. **InfluxDB → Dashboard Cache (Redis)** (latency: < 50ms)

## 5. Redis → User Dashboard (latency: < 30ms)

**Total end-to-end latency from device reading to dashboard display: < 430ms** under normal load.

The pipeline processes a peak throughput of **2.4 million data points per minute** across all connected devices.

---

## 9.4 Platform FAQ

**Q: A Business Plan user experiences 2 hours and 30 minutes of downtime beyond their SLA limit.**

**What credit do they receive?** Breach = 2.5 hours. Tier = 15%. Credit = ₹4,999 × 0.15 = **₹749.85**

**Q: A Pro user makes 75 API calls in 1 minute. What happens to the excess 15 calls?** The first 60 calls succeed. The 15 excess calls receive **HTTP 429 Too Many Requests** with a Retry-After header.

**Q: What is the maximum end-to-end latency from a smart meter reading to it appearing on the dashboard?** < 430 milliseconds under normal load.

---

## PAGE 10 — Master FAQ & Numerical Problem Set

### 10.0 Comprehensive Numerical Questions

This section consolidates value-based questions across all topics for quick reference. These are the kinds of queries VoltBot is designed to answer using data from this knowledge base.

---

### Section A — Billing & Subscriptions

**Q1: A user is on the Pro Plan (monthly) and upgrades to Business Plan on Day 12 of a 30-day cycle. What is the prorated charge?**

- Pro daily rate = ₹1,299 ÷ 30 = ₹43.30/day
- Business daily rate = ₹4,999 ÷ 30 = ₹166.63/day
- Remaining days = 18
- Prorated charge = (₹166.63 – ₹43.30) × 18 = ₹123.33 × 18 = **₹2,219.94**

**Q2: A Business Plan user cancels their annual subscription after 7 months. How much is refunded?**

- Annual fee = ₹47,990
- Monthly equivalent = ₹47,990 ÷ 12 = ₹3,999.17
- Remaining months = 5
- Refund = ₹3,999.17 × 5 = **₹19,995.83**

**Q3: A Business account has 134 devices active. What is the total monthly invoice before GST?**

- Base = ₹4,999

- Overage =  $(134 - 100) \times ₹49 = 34 \times ₹49 = ₹1,666$
- Total before GST = **₹6,665**

## Section B — Energy Consumption

**Q4: A household used 180 kWh in April and 230 kWh in May. What is the MoM change percentage?**  $((230 - 180) \div 180) \times 100 = +27.78\%$  increase

**Q5: Using TSSPDCL slab rates (₹2.35 / ₹4.80 / ₹6.90 / ₹8.50), what is the bill for 420 kWh consumption?**

- Slab 1:  $100 \times ₹2.35 = ₹235$
- Slab 2:  $100 \times ₹4.80 = ₹480$
- Slab 3:  $100 \times ₹6.90 = ₹690$
- Slab 4:  $120 \times ₹8.50 = ₹1,020$
- **Total energy charge = ₹2,425** (add fixed charges separately)

**Q6: An AC (2.5 kW) runs 9 hours/day. What is the monthly consumption and cost at ₹7.50/unit?** Daily =  $2.5 \times 9 = 22.5$  kWh. Monthly =  $22.5 \times 30 = 675$  kWh. Cost =  $675 \times ₹7.50 = ₹5,062.50$

## Section C — Solar

**Q7: A solar system generates 290 kWh, self-consumes 210 kWh, exports 80 kWh, and the household imports 160 kWh. What are net billed units and the bill at ₹6.90/unit?** Net units =  $160 - 80 = 80$  kWh. Bill =  $80 \times ₹6.90 = ₹552$

**Q8: A 4 kW system costs ₹2,40,000 and generates 16 kWh/day. At ₹8/unit, what is the payback period?** Annual generation =  $16 \times 365 = 5,840$  kWh. Annual saving =  $5,840 \times ₹8 = ₹46,720$ . Payback =  $₹2,40,000 \div ₹46,720 = 5.14$  years

## Section D — Peak Hours & Savings

**Q9: A household uses 20 kWh/day. 35% falls in peak hours (₹11/unit) and 65% off-peak (₹6/unit). What is the daily cost, and how much is saved monthly by shifting all peak consumption to off-peak?** Current:  $(20 \times 0.35 \times ₹11) + (20 \times 0.65 \times ₹6) = ₹77 + ₹78 = ₹155/\text{day}$  All off-peak:  $20 \times ₹6 = ₹120/\text{day}$ . Daily saving = ₹35. **Monthly saving = ₹35 × 30 = ₹1,050**

**Q10: An AC (1.5 kW, 10 hrs/day) currently runs at 18°C. What is monthly consumption, cost at ₹8/unit, and saving if temperature is raised to 26°C?** Monthly =  $1.5 \times 10 \times 30 = 450$  kWh. Cost =  $450 \times ₹8 = ₹3,600$  Temperature increase =  $8^\circ\text{C}$ . Savings =  $8 \times 6\% = 48\%$ . New cost =  $₹3,600 \times 0.52 = ₹1,872$ . **Monthly saving = ₹1,728**

## Section E — Carbon Footprint

**Q11: A commercial facility consumes 4,500 kWh/month. What is the annual CO<sub>2</sub> footprint at 0.82 kg/kWh?** Monthly CO<sub>2</sub> =  $4,500 \times 0.82 = 3,690$  kg. Annual =  $3,690 \times 12 = 44,280$  kg = **44.28 tonnes CO<sub>2</sub>/year**

**Q12: If the above facility reduces consumption by 18% through VoltStream recommendations, how many kg CO<sub>2</sub> is saved annually?** Reduction =  $4,500 \times 0.18 = 810$  kWh/month. CO<sub>2</sub> saved =  $810 \times 0.82 \times 12 = 7,970.4$  kg/year = **7.97 tonnes**

---

## Section F — VoltBot & Platform

**Q13: A Business Plan user experienced 3 hours 20 minutes of excess downtime this month. What is their SLA credit?** 3h 20min = 3.33 hours. Tier = 15% (1–4 hours). Credit =  $₹4,999 \times 0.15 = ₹749.85$

**Q14: A Pro user sends 380 SMS alerts in a month. What is the SMS overage charge?** Free quota = 200. Overage =  $180 \times ₹0.15 = ₹27.00$

**Q15: A Pro user has 12 devices. How many data points are generated per month (30 days) at 1-minute intervals?**  $1,440 \times 12 \times 30 = 518,400$  data points/month

---

*End of VoltStream Complete Knowledge Base Documentation*

v3.0.1 Support: [support@voltstream.io](mailto:support@voltstream.io) | Billing: [billing.voltstream.io](mailto:billing.voltstream.io) | Status: [status.voltstream.io](https://status.voltstream.io) ©

2026 VoltStream Technologies Pvt. Ltd. All Rights Reserved.

---

That's your complete single PDF — **10 pages**, covering all 6 topics with theory, real numerical examples, formulas, tables, and a full **15-question master problem set** on Page 10 that VoltBot can answer precisely using RAG retrieval. Ready to convert to PDF whenever you are!